Semester: 6th
Subject Name:- Cloud Computing & Code:CS30010
Branch (s): CSE

SPRINGMID SEMESTER EXAMINATION-2026

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
Subject Name
[Subject Code]

Time: 1 1/2 Hours **Full Mark: 20**

Answer Any four questions including question No.1 which is compulsory.
The figures in the margin indicate full marks.
Candidates are required to give theiranswers in their own words as far as practicable and all parts of a question should beanswered at one place only.

1. Answer all the questions. [1Mark X5]

   a) Briefly describe two popular Hypervisor used in virtualization.
   b) What are some common challenges that organizations face when adopting cloud computing, and how can they be addressed?
   c) What do you mean by Utility Computing? Explain few chareterstics.
   d) What do you mean by Service Orchestration. Mention the name of layers associated with it.
   e) Why multi-tenancy is not considered as an essential feature of cloud computing? Mention briefly.

2.A mid-sized financial services firm currently operates entirely on a Private Cloud to meet strict regulatory data sovereignty requirements. However, they are launching a new AI-driven customer analytics platform that requires massive, unpredictable bursts of computational power and access to proprietary machine learning APIs available only on hyperscale providers (like AWS or Azure).

(i) Identify and justify the most appropriate cloud deployment model for this transition.

(ii) Analyze the trade-offs regarding security, cost-efficiency, and operational complexity compared to their current Private Cloud setup.

(iii) Propose a solution for how the firm can handle 'Cloud Bursting' within this model without violating data residency laws."[5Marks]

3. Resource abstraction through virtualization: What does this mean? Explain the working of Hardware-Assisted Virtualization. [5 Marks]

4. Compare and contrast the three main cloud service models (IaaS, PaaS, and SaaS) with respect to business contexts. Give examples of each, and explain the advantages and disadvantages of each model. [5 Marks]

5. Explain the concept of elasticity in cloud computing and how it is different from scalability. Give a comparison between Cluster, Grid and Cloud Computing.

[5 Marks]

*** Best of Luck ***